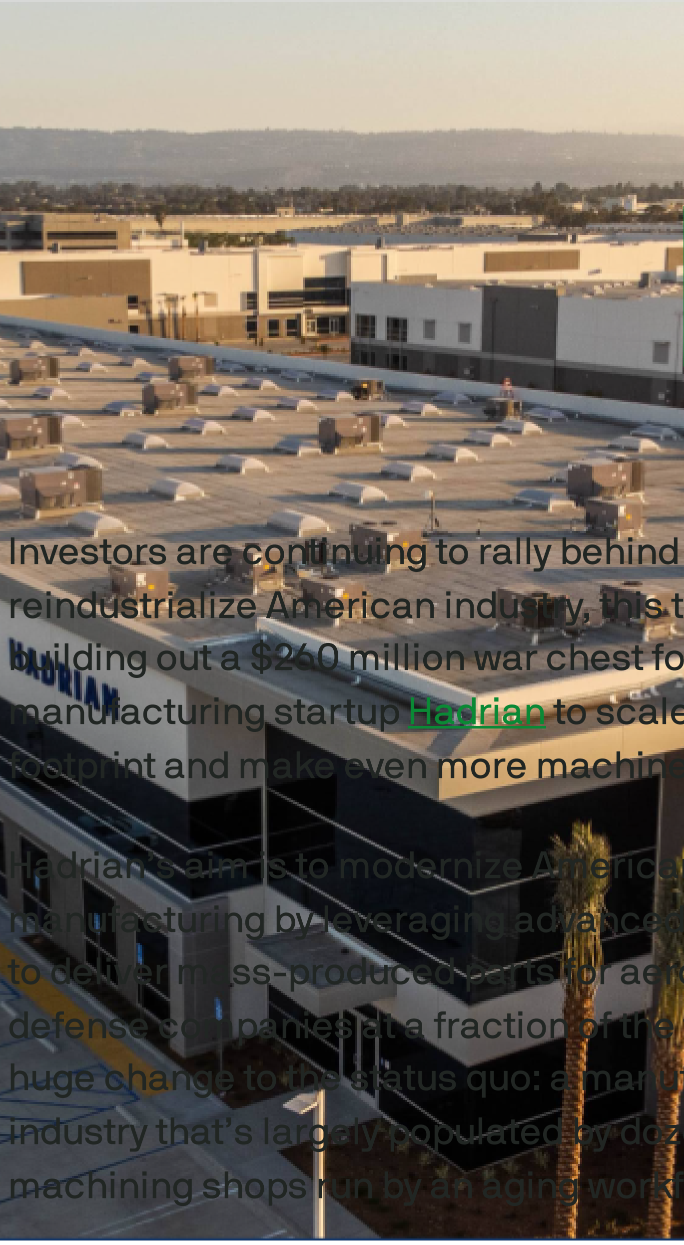
STARTUPS

Hadrian raises $260M to build out automated factories for space and defense parts

Aria Alamalhodaei — 11:16 AM PDT · July 17, 2025



Investors are continuing to rally behind the call to reindustrialize American industry, this time by building out a $260 million war chest for automated manufacturing startup Hadrian to scale its factory footprint and make even more machine parts.

Hadrian's aim is to modernize American manufacturing by leveraging advanced automation to deliver mass-produced parts for aerospace and defense companies at a fraction of the time. It's a huge change to the status quo: a manufacturing industry that's largely populated by dozens of small machining shops run by an aging workforce.

IMAGE CREDITS: HADRIAN

Hadrian's first target was high-precision CNC machining, a manufacturing process that makes parts to extremely tight tolerances — often measured in the microns, not millimeters (a single human hair is anywhere from 50 to 120 microns in thickness). Now, in addition to that core CNC offering, the company is getting ready to diversify into welding, casting, additive, and other processes, Hadrian founder and CEO Chris Power said in a post on X. (He did not immediately respond to TechCrunch's request for comment on the new round.)

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Chris Power
@2112Power · Follow

Hadrian is announcing a $260m Series C lead by @foundersfund + @Lux_Capital + @MorganStanley for (Factory Expansion financing). We're also welcoming new investors @altimeter, @1789Capital + investments from existing investors @a16z, @constructcap, @137ventures and many others. Show more

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The hefty new round will also go toward building out a new Arizona facility, dubbed "Factory 3," slated to come online by Christmas 2025. That factory will offer four times the machining throughput of Hadrian's second factory. Hadrian is also expanding its 500,000-square-foot headquarters and R&D space in Torrance, California, with the new funding.

The company will also start offering divisions for maritime and munitions-specific parts "to meet the sale and speed needed to reclaim our birthright as the industrial superpower of the world," Power said on X.

Hadrian's business model is not just selling parts, but also a "factories as a service" model that will see dedicated facilities for customers that want to ensure committed factory capacity.

Speaking at the Reindustrialization Summit on Wednesday, Power argued that reshoring domestic production is nothing less than existential: “This country is heading into a generational fight,” he said. “The hour is extremely late. The great game is on … We have an incredibly short window to prepare for this, fix it, reindustrialize the country and return to what made us great in the first place.”

The new round was led by Founders Fund and Lux Capital, with Morgan Stanley providing financing for the factory expansion. New investors Altimeter and 1789 Capital, and existing investors a16z, Construct Capital, and 137 Ventures also participated. The company has now raised nearly $500 million since it was founded in 2020.

Topics: Chris Power Founders Fund Hadrian lux capital Space Startups

the founders fund

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Hadrian is building the factories of the future for rocket ships and advanced manufacturing

Jonathan Shieber 9:41 AM PDT · April 15, 2021

IMAGE CREDITS: NASA / GETTY IMAGES

If the eight person team behind the new startup Hadrian has their way, they'll have transformed the manufacturing industry within the next decade.

At least, that's the goal for the new San Francisco-based startup, founded only last year, which has set its sights on building out a new model for advanced manufacturing to enable the satellite, space ship, and advanced energy technology companies to build the future they envision better and faster.

"We view our job as to provide the world's most efficient space and defense component factory," said Hadrian founder, Chris Power.

Initially, the company is building factories to make the parts that go on rocket ships, according to Power, but the business has implications for any company that needs bespoke components to make their equipment.

"Let me tell you how bad it is at the moment and what's going to happen over the next 20 years. Right now everyone in space and defense, [including] SpaceX and Lockheed Martin, outsources their parts and manufacturing to small factories across the country. They're super expensive, they're unreliable and they're completely invisible to the customers," said Power. "This causes big problems with space and defense manufacturers in the design phase, because the lead time is so long and the iteration time is super long. Imagine running software and being able to iterate on your product once every 20 days? If you can imagine a Gantt chart of how to build a rocket, about 60% of that is buffer time… A lot of the delays in launches and stuff like that happen because parts got delivered three months ago. It'd be like running a McDonalds and realizing that your fries and burger providers could not tell you when the food would arrive."

It’s hard to overstate the strategic importance of the parts suppliers to the operations of aerospace, defense, and advanced machining companies. As no less an authority on manufacturing than Elon Musk noted in a tweet, “The factory is the product.” It’s also hard to overstate the geopolitical importance of re-establishing the U.S. as a center of manufacturing excellence, according to Hadrian’s investors Lux Capital, Founders Fund, and Construct Capital. Which is one reason why they’re investing $9.5 million into the very early stage business.

“America made massive strategic mistakes in the early 90s which have left our national manufacturing ecosystem completely dilapidated,” said Founders Fund principal Delian Asparouhov. “The only way to get out of this disaster is to re-invent the most basic input into our aerospace and defense supply chains, machining metal parts quickly and with high tolerance. Right now

The factory is the product

— Elon Musk (@elonmusk) January 11, 2021

Power got to understand the problem at his previous company, Ento, which sold workforce management software to blue collar customers. It was there he realized the issue of. the aging workforce and the need for manufacturers to upgrade almost every aspect of their own technology stack. “I realized that the right way to bring technology to the industrial space is not to sell software to these companies, it’s to build an industrial business from scratch with software.”

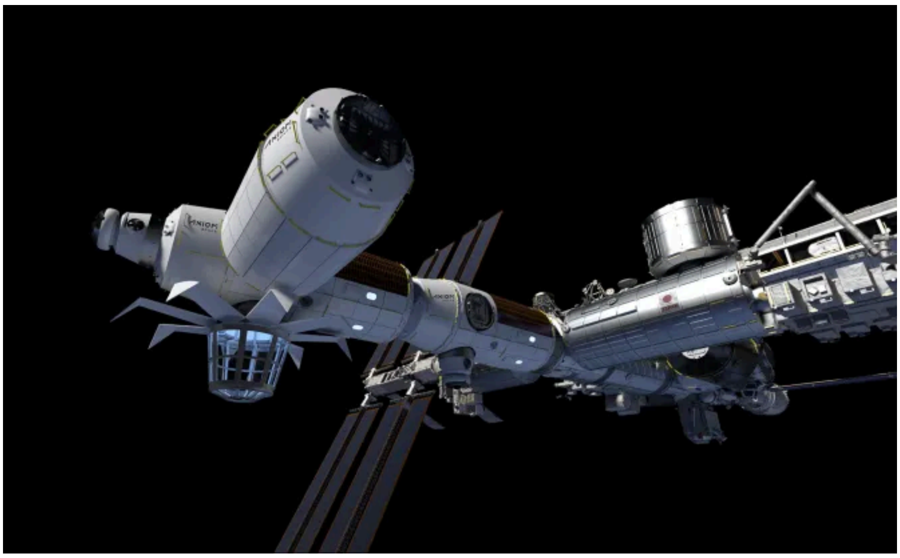
Initially, Hadrian is focusing all of its efforts on the space industry, where the component manufacturing problem is especially acute, but the manufacturing capabilities the company is building out have broad relevance across any industry that requires highly engineered components.

“The demand for manufacturing from both the large SpaceX and Blue Origin all the way to this growing long tail of companies from Anduril to Relativity to Varda,” said Lux Capital co-founder Josh Wolfe. “Most of these guys are using mom and pop machine shops… [and] those shops are horribly inefficient. They’re not consistent, and they’re not reliable. Between the software automation, the hardware, you can cut down on inefficiency every step of the process… I like to think of value creation as waste reduction… so mundane things like quoting, scheduling, bidding, and planning all the way to the programming of the manufacturing… every one of those things takes hours to tens of hours to days and weeks, so if you can do that in minutes, it’s just a no-brainer. [Hadrian] will be the cutting edge choice for all of the new and explicitly dedicated and focused aerospace and defense companies.”

Power envisions a network of manufacturing facilities that can initially cover roughly 65% of all space and defense components, and will eventually take that number up to 95% of components. Already several of the biggest launch vehicle and satellite manufacturers are in talks with the company to produce hundreds of units for them, Power said. Some of those companies just happen to be in the Construct, Lux, and Founders Fund portfolio.

And the company’s founder sees this as a new way to revitalize American manufacturing jobs as well. “Manufacturing jobs in space and defense can easily be as high paying as a software engineering job at Google,” he said. In an ideal world, Hadrian would like to offer an onramp to high paying manufacturing careers in the 21st century in the same way that automakers provided good union jobs in the twentieth.

“We haven’t built any of this. If you look at the sheer number of people that we need to train and hire on our new technology and new systems, that people problem and that training problem is part of growing our business.”



A RENDER OF AXIOM'S FUTURE COMMERCIAL SPACE STATION DESIGN.

Topics: aerospace america Blue Origin Co-founder Construct Capital Elon Musk entrepreneurship factory food Founders Fund Google Josh Wolfe Lockheed Martin lux capital manufacturing mcdonalds San Francisco software automation SpaceX Startup company TC United States

Jonathan Shieber

Writer

Jonathan was an editor at TechCrunch./p>

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Varda Space Industries closes $42M Series A for off-planet manufacturing

Aria Alamalhodaei 2:40 PM PDT · July 29, 2021

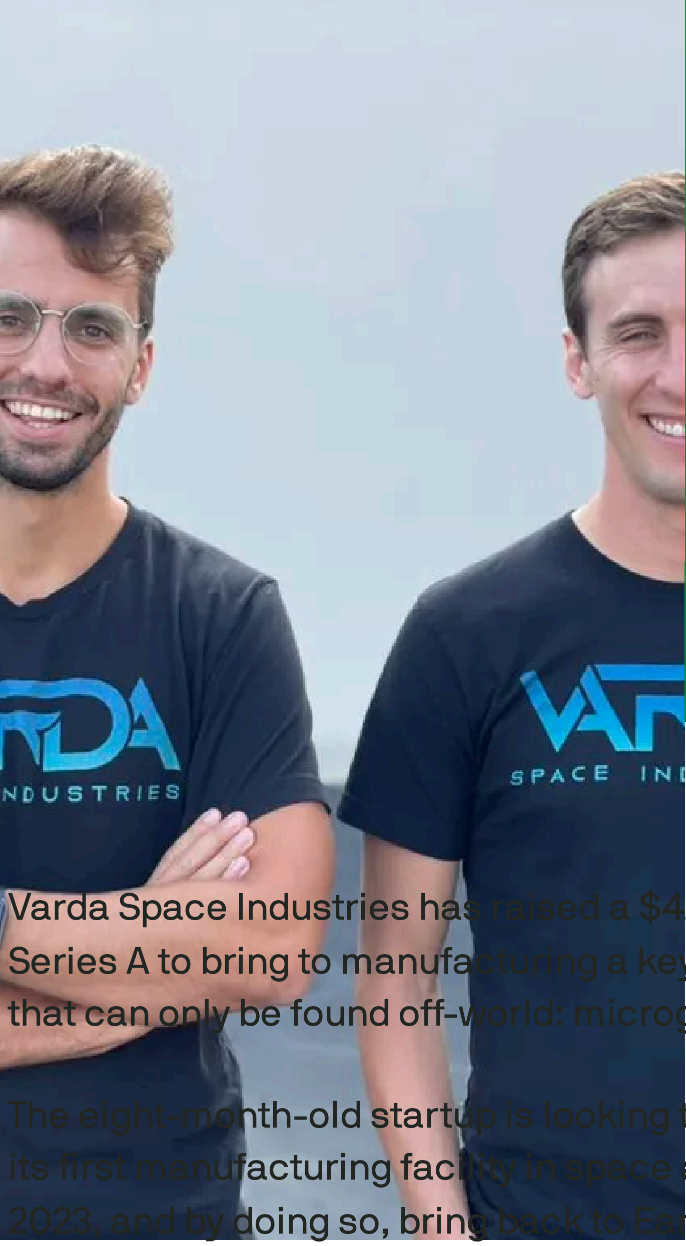

IMAGE CREDITS: VARDA SPACE INDUSTRIES

Varda Space Industries has raised a $42 million Series A to bring to manufacturing a key capability that can only be found off-world: microgravity.

The eight-month-old startup is looking to establish its first manufacturing facility in space as early as 2023, and by doing so, bring back to Earth advanced products that can only be made under sustained periods of zero gravity.

The round was led by Khosla Ventures and Caffeinated Capital, with participation from existing investors Lux Capital, General Catalyst and Founders Fund. It pushes the company's total raise thus far to over $50 million, including a $9 million seed round last December.

Varda's idea is different than that of Jeff Bezos, who said after his own trip to space earlier this month that he wants to "move all heavy industry and all polluting industry off Earth." The company's co-founders, SpaceX veteran Will Bruey and Founders Fund principal Delian Asparouhov, aren't imagining cement mixers and steel plants in orbit. Instead, they want to open up manufacturing processes that

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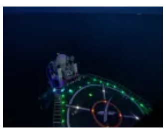

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Aria Alamalhodaei

- Jul 30, 2025

aren't possible on Earth, in order to make bioprinted organs, fiber-optic cables or pharmaceuticals — products that require fundamentally different conditions than what's available on-planet.

Building the space factory of the future

The value of microgravity manufacturing, Bruey and Asparouhov say, can be found with the International Space Station, essentially a scientific outpost. A steady stream of research has emerged from the ISS over the last few decades showing that novel materials and products are possible in space. But until now, getting, staying in and returning from orbit has been too costly to consider scaling these findings.

"In a way, a lot of our R&D has already been done for us in the public sector, and we're essentially a ramp toward commercialization for that research that's already been proven out," Bruey told TechCrunch.

Space manufacturing startup Varda, incubated at Founders Fund, emerges with $9 million in funding

From a young age, Will Bruey, the co-founder and chief executive of Varda Space Industries, was fascinated with space and running his own business. So when the former SpaceX engineer was tapped by Delian Asparouhov and Trae Stephens of Founders Fund to work on Varda, he didn't think twice. Bruey spent six years at SpaceX ... Continue reading

Right now, the company is building a three-module spacecraft comprised of an off-the-shelf satellite platform, a center platform where the microgravity manufacturing will take place, and a reentry vehicle to bring the materials back to Earth. For the first 10 or so launches, Bruey said Varda would build the products itself. Once the company has established that its process is reliable and cheap, he added that in the long term the goal is to become a contract manufacturing platform for other companies wanting to build products in space.

Asparouhov likened it to the iPhone and the App Store: "The iPhone didn't come out with the App Store. Apple developed the first 10 or 11 apps to share the value of that. So we're developing those first few apps ourselves to show the value in this commercial capability that we're bringing to market, but over time, we will start to release an app store."

One key part of Varda's plan is to make all of the manufacturing automated. By keeping humans out of the picture (at least for now), the company is able to reduce critical overhead by skipping humanrated spacecraft development (and the associated safety concerns with crewed launches).

VARDA INVITED REGULATORS AND THE DOD TO A PRELIMINARY DESIGN REVIEW.

IMAGE CREDITS: VARDA SPACE INDUSTRIES

"I think that what investors, NASA and the [Department of Defense], really see as exciting about our approach is that in comparison to everyone else that's ever discussed 'space manufacturing,' we're by far the most near-term, pragmatic, commercially viable approach, launching and producing materials less than 18 months from now, as opposed to plans that are typically five years, 10 years, decades away from being viable," Asparouhov said.

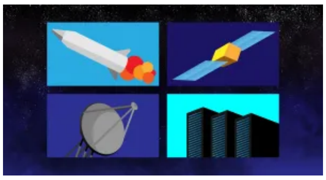
He added that one way to think about space manufacturing is that there is a certain dollar per unit-mass that Varda will need to spend to get things up to microgravity, and a dollar per unit-mass of value from manufacturing in microgravity. The key to profitability is finding the products that maximizes the difference between these two equations. Novel pharmaceuticals, for example, could yield massive profits if the innovation gains from zero gravity are correspondingly high.

The company is imagining "multiple missions" in 2023, Bruey said, and then moving to once per quarter and even imagining multiple reentry capsules returning with products per day. The Varda co-founders are convinced that the scale of demand for novel space-made products is potentially high enough to meet this kind of launch and reentry schedule.

Compared to a burgeoning industry like space tourism, Bruey said the space manufacturing has the potential to positively affect a much higher portion of humanity.

"It will touch many different parts of humanity's experience out here on Earth, with significant improvements in quality of life," he said.

As launch market matures, space opportunities on the ground take off



The space economy in the last few years has been in large part driven by the increasing cadence and reliability of launch services, and while that market will continue to grow, the new economy enabled by those launches is only just beginning to take off. If you thought the launch boom was big, just wait … Continue reading

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